In Search of Mirror Neutrons: Prototyping a Tri-Helical Coil with a Longitudinally Tunable Uniform Omnidirectional Field

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January 29, 2026

1 Introduction

1.1 Background

An unanswered question in particle physics is the discrepancy between the lifetime of the neutron as measured by two different experimental methods: the "Bottle" Method and the "Beam" Method. The "Bottle" Method entails trapping a certain number of ultracold neutrons in a secure container and measuring the number of neutrons still in the container as a function of time, thereby measuring the decay rate and neutron lifetime. Alternatively, the "Beam" Method measures the rate of appearance of protons, which neutrons decay into. By measuring the number of protons produced by a "Beam" of cold neutrons as a function of time, the lifetime of the neutron can be determined. The problem arises when, after accounting for any systemic error in these experimental methods, there remains a discrepancy between the results of about nine seconds with $\pm$ two seconds of uncertainty.

Mirror neutrons have been proposed as an explanation for the difference between these experimental results. Mirror matter is a theorized category of matter composed of hypothetical particles that are thought to have similar yet mirrored properties to the ordinary particles of the Standard Model. They have no mechanism for interacting with ordinary matter except through the gravitational force.[6] It has been proposed that the discrepancy in the measured lifetime of the neutron is due to systematic oscillations of ordinary neutrons into mirror neutrons in the presence of a magnetic field. Once a neutron becomes a mirror neutron, it is no longer detectable and seems to disappear. Should evidence for mirror matter be discovered, it may also provide an explanation for so called "Dark Matter", which is another unsolved puzzle in the Standard Model.[4]

1.2 Main Goal/Hypothesis

We hypothesize that Mirror Neutrons exist, and our goal is to detect them by exposing a beam of neutrons to a specific and uniform magnetic field to induce oscillation to the mirror state. Once exposed to the magnetic field, the beam will pass through a cadmium absorber that will prevent the passage of non-mirror particles. Any neutrons that passed through the barriers and were detected will necessarily have oscillated from ordinary to mirror and back to ordinary neutrons and therefore provide evidence for the existence of mirror matter.

For this experimental set-up, we will require the ability to produce a uniform magnetic field along the path of the neutron beam that can be oriented in any direction. This is necessary because before neutrons will oscillate between ordinary and mirror states, both the ordinary and mirror ambient B-fields, which are not necessarily aligned, must be equal in magnitude and orientation.[6] To produce such a B-field we have designed and built a novel tri-helical coil.

Our coil design is composed of three homogeneous solenoids layered one on top of the other that are slanted at 30° and clocked by 120° each. Additionally, we incorporated Digital Analog Converters (DAC) into the coil design which moderate the current from 0 mA to 120 mA for every winding pair, creating twenty-one sub-coils for every coil. We can exploit this coil configuration to generate a B-field in any direction independent of the coil's physical orientation by manipulating the relative current strength in the three component coils. Furthermore, selecting specific current distributions for the sub-coils allows us to increase the uniformity of the B-field inside of the coil.

2 Acknowledgments

At the time I took charge of the project, the previous owner, Andrew Winterman, had already reached several important milestones. In particular, Andrew's designs for both the flex PCB coil traces and the CAD support cylinder file continued to be part of the coil design throughout the project. Details on Andrew's design process for both can be found in his 2024 REU Report.[9] Furthermore, at the beginning of the project, Winterman provided support by explaining the work done so far, and by giving advice on what immediate next steps should be taken.

Greg Porter of the University of Kentucky is the designer of the DAC/Multiplexer boards and the code to operate them. He also explained to me how the boards and the code he wrote for the board components function so that I could operate the coil once I had built it.

This work was supported by the U.S. Department of Energy under contract DE-SC0014622 and by the UK College of Arts and Sciences SURA program.

This project was conducted as part of the nn' Collaboration.

Lastly, none of this would have been possible without the help of my faculty mentor, Dr. Christopher Crawford, who provided the overall vision for the project and assistance throughout.

3 Important Components

In this section, I describe the major components of the tri-helical coil.

3.1 Flex PCB

The windings of the coil are made up of three flex PCBs of the design shown below.

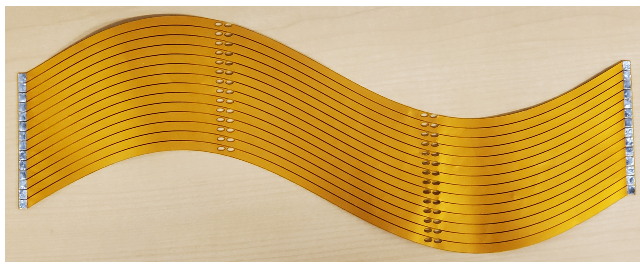


Figure 1: Andrew's flex PCB.

There are fourteen copper traces coated on both sides with and separated by insulating material. At both ends of the PCB are segmented solder pads. The copper trace inner section has the shape of a sine wave, with an amplitude and wavelength such that when the flex PCB is wrapped around a cylinder the slant of the coil loop is approximately 30°. More information on the design of the flex PCBs can be found in Andrew's 2024 REU Report.[9]

3.2 DAC/Multiplexer Board

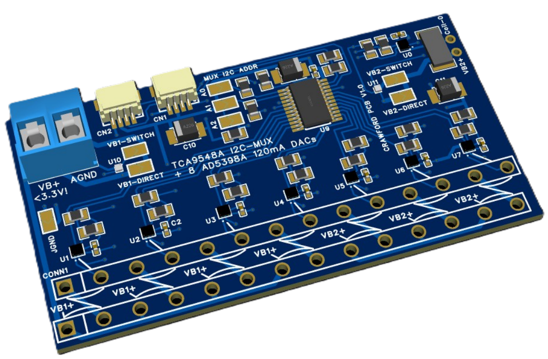


Figure 2: DAC/Multiplexer circuit board designed by Greg Porter.

The DAC/Multiplexer circuit board's main purpose is to control the current in the fourteen coil windings that are connected through to the power supply it via seven AD5398A 120 mA, current sinking, 10-Bit, I2C DAC chips.[2] A TCA9548 8-channel Multiplexer is included in the circuit board to enable communication over I2C with individual DACs, as each DAC shares the same I2C address.[5] The secondary purpose of the DAC/Multiplexer boards is to provide voltage from the power supply to the coil windings. By using seven DACs per board, I can specify the current from 0 mA to 120 mA in each of the sub-coils.

3.3 Power Source

The use of the DACs in the system puts some limitations on what can be used to power the coil. The DACs can only work with DC power, and because there are no AC to DC power converters in the circuit boards, the power source then must be DC. Additionally, any voltage over 3.0 V risks damaging the DACs, so the power source should be at some voltage below that, but above 1.0 V.

3.4 Adafruit QT2040 Trinkey

The QT2040 is an Adafruit RP2040 microcontroller mounted to a USB A key with a STEMMA QT I2C port for communication with other micro devices.[8] It can run both MicroPython and CircuitPython programs, and is the component through which I send commands to the DAC/Multiplexer boards to control the sub-coil currents.

4 Building the Variable Current Tri-coil

In this section, I outline the process that I followed to construct the tri-helical coil and power supply.

4.1 Support Cylinder

Before I could construct the coil, I first needed to fabricate a cylindrical component to support the coil and ensure that the flex PCBs conformed to a cylindrical shape as closely as possible. Andrew Winterman had already designed a suitable CAD file for such a cylinder during his time on the project, and was he was kind enough to share it with me.[9]

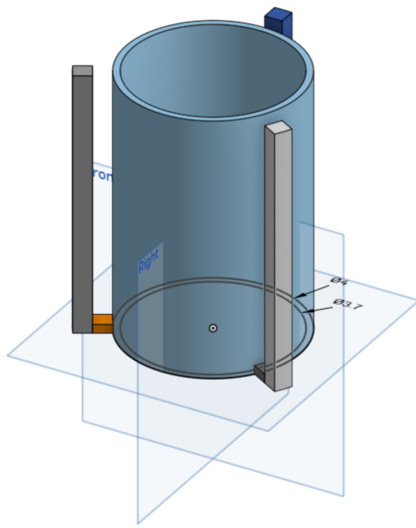


Figure 3: Andrew's cylinder CAD design.

There are only three aspects of the cylinder that need to be considered in the design process: the outer diameter, the height, and the magnetic properties of the fabrication material. The outer diameter of the cylinder should be four inches, as any more and the separation of solder pads on the outermost coil will be too great to accommodate the header pins, and any less and the solder pads of the innermost coil will overlap too much. The height of the cylinder must be at least 13 centimeters to support the full length of the coil. Lastly, the cylinder should be made with a material that will not interfere with the coil's B-field in any way. For my support cylinder, I used Dr. Crawford's Creality CR-10 3D printer with PLA filament.

4.2 Mounting the Flex PCBs

With the support cylinder printed, my next task was to mount the flex PCBs around the cylinder as flush and as well aligned as possible. The flex PCBs are designed such that when wrapped around a cylinder, the solder pads on each end should meet and sit one pad beside its correspondent, so that all of the pads together form a rectangular grid. Keeping this in mind, my method for mounting the flex PCBs was as follows.

I wrapped one of the PCBs around the cylinder, focusing only on keeping the solder pads aligned and getting the PCB wrapped as tightly as possible around the cylinder. While holding the PCB in place by hand, I recruited a second person to apply tape, joining the two ends and securing the PCB. When using a four inch diameter cylinder, it may be necessary to apply a strip of insulation to the underside of one of the solder pad strips on the innermost coil to avoid shorting of the circuit, as there is some overlap. With the inner coil wrapped, I slid it up the cylinder until the top edge was flush with the edge of the cylinder. I did this to make aligning the other two coils easier. Once the PCB was in position, I secured it to the cylinder with Scotch Tape™.

The process for mounting the middle and outer coils was mostly the same as the inner coil; however, there was an additional concern in that it was also necessary to align the drill holes in the outer coils with the solder pads of the inner coils. This means that the positioning of the outer two coils is largely determined by the position of the inner coil.

4.3 Soldering the Header Pins

The chips that contain the DACs are designed to connect to the flex PCBs via header pins, so my next step was to solder said header pins to the eighty-four solder pads. The header pins I used were 24 Position Terminal Block Header, Male Pins, Unshrouded 0.197" (5.00 mm) Vertical Through Hole from Würth Elektronik.[3] When soldering the header pins, it was important that the pins were properly aligned to allow the chips to be attached later and that the solder joints made a good connection between the pins and the solder pads. To keep the pins in the right position while soldering, I used a blank of the DAC chips as a jig.

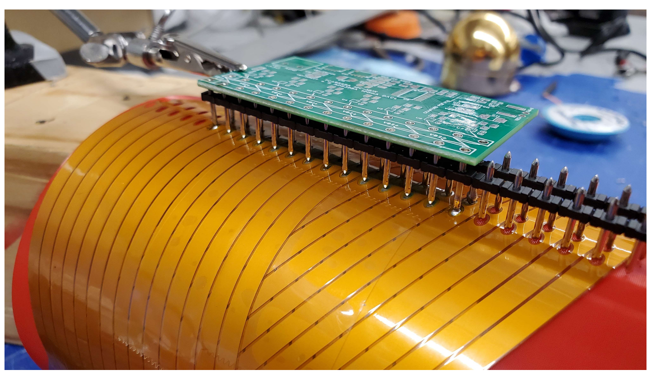


Figure 4: Jig setup used to hold the header pins in place.

With the pins held in place, I then soldered the four pins at the top and bottom. As only fourteen pins are needed for the coil, the twenty-four pin header block should be trimmed down before soldering. After the end pins were soldered and holding the other pins in place, I removed the jig and soldered the rest. This, however, was not the optimal method, because without the jig the pins would become misaligned as I soldered them. The correct way to attach the pins would be to solder the corner pins as I did, and then keeping the jig in place, solder all of the pins on one side. Then flip the jig around so that the other set of pins are exposed and solder the rest.

One difficulty with the method that I followed was that it required me to solder some of the pins to the inner solder pads through the drill holes of the outer coils. Because of this, I recommend using a solder iron with a fine tip, as it is important to heat both the pin and the solder pad sufficiently to get a good bond with the solder. If either the pin or the solder pad is not hot enough, this could result in a cold solder joint where the solder does not properly bond to one of the surfaces. This will cause a poor electrical connection and should be avoided.

Once all of the header pins were soldered to the coils, I used a multimeter to verify that there were no shorts between coil windings. I also measured the resistance through the turns to verify that the pins had a good connection to the solder pads.

Figure 5: Mounted tri-helical coil with pins attached.

4.4 Attaching and Wiring DAC/Multiplexer Boards

Soldering the DAC/Multiplexer chips to the coil at this point was straightforward. It did not matter which way the chips were oriented on the coil, only that they all shared the same orientation. I was also careful to ensure that all of the solder joints made good connections to the pins and the pin holes in the chips.

All of the chips have both an in and an out I2C port, and I wired them together in series via STEMMA QT 200 mm 4-pin cables. To the remaining in I2C port I connected via a STEMMA QT 400 mm 4-pin cable the Adafruit QT2040 Trinkey, and the remaining out I2C port is available to attach additional devices to in the future.

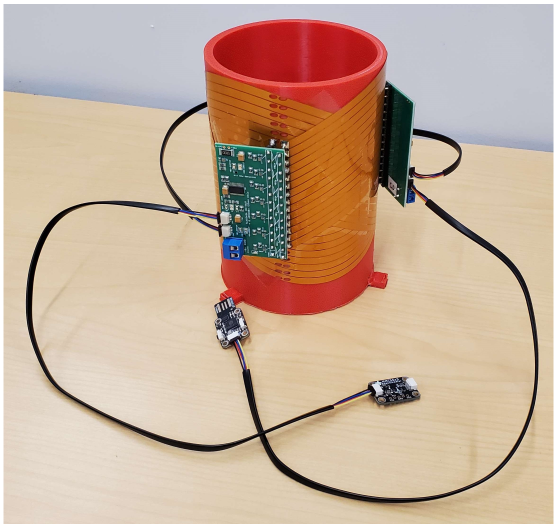


Figure 6: Tri-helical coil with DAC/Multiplexer chips.

4.5 Creating the Support Cradles

In order to map the field of the coil, it was necessary to be able to set it stably on its side. Therefore, using the CAD software Onshape™, I designed two support rectangles with circular sections cut out of the same diameter as the cylinder.

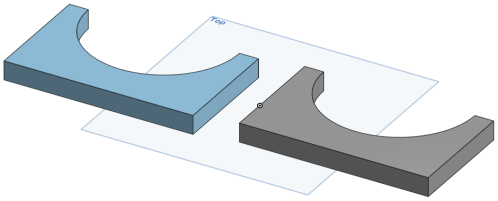


Figure 7: Support cradle CAD design.

I then printed these with the same Creality CR-10 that I used to print the cylinder, and after removing some material from one of the supports with a rounded file, I secured the supports to the cylinder with hot glue.

5 Building the DC Power Supply

When building the power supply, I had two goals: generate no more than 1.5 V and be able to run the coil at full power (2.52 A) for a reasonable amount of time before the voltage dropped below 1 V. I choose to use five Procell®Constant D PC1300 batteries wired in parallel.[7] By wiring in parallel, the voltage remains capped at 1.5 V and according to the datasheet, five batteries will run at 2500 mA for about 18 hours before the voltage drops below 1 V.

To make the battery pack, I hot glued five D-Cell battery holsters to a piece of 2x4 lumber, and connected all of the positive and negative terminals together with short pieces of stranded wire and solder. I included a rocker switch in the circuit between the positive terminal and the coil. Finally, to connect the coil, I soldered two long wires to the rocker switch and the negative terminal. These wires were crossed over about a foot from their end by a two-foot piece of wire and soldered together so that the coils could be wired in parallel as well.

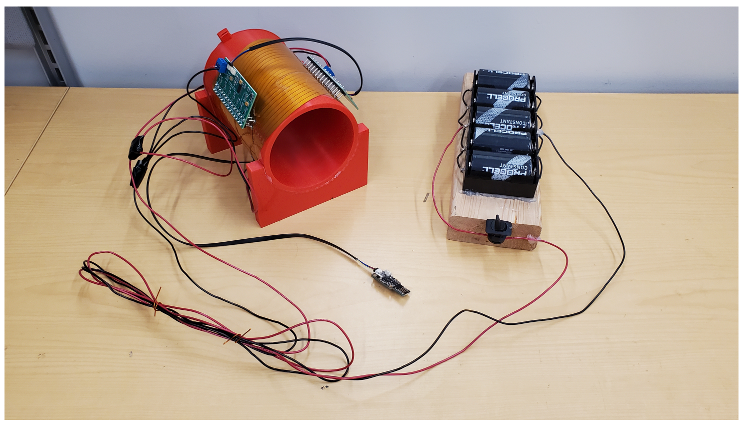


Figure 8: Finished tri-helical coil and 1.5 V power supply.

6 Verifying the Coil

6.1 Auto-Mapper and Mu-metal Shield

With the coil fully constructed, the next step was to map out the B-field that each sub-coil generated to verify that the coil was working properly. To do this I used both the auto-mapper and the Mu-metal shield from Dr. Crawford's lab.

The auto-mapper consists of a Bartington Mag-13MS1000 three-axis flux-gate magnetometer mounted to a three-axis gantry system driven with stepper motors.

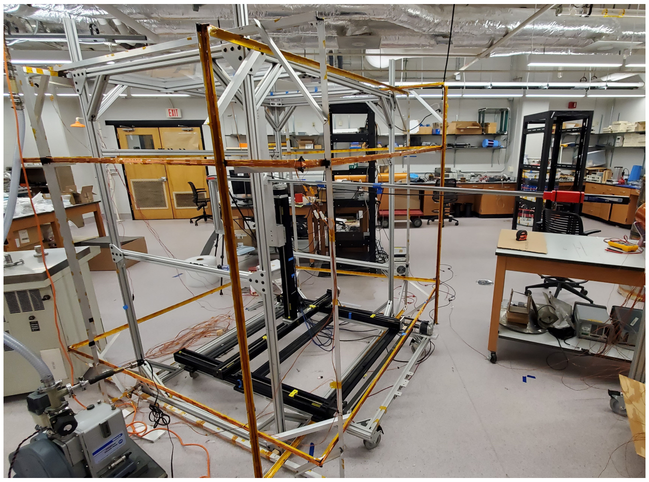
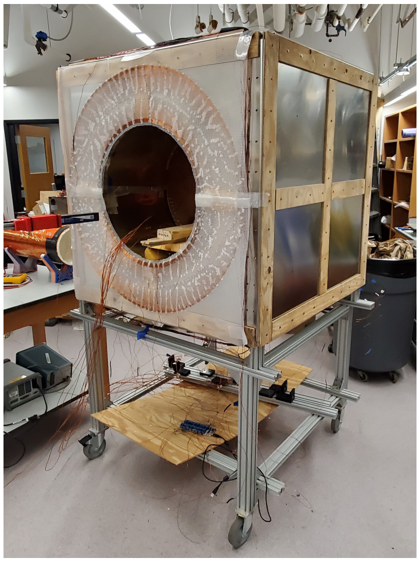


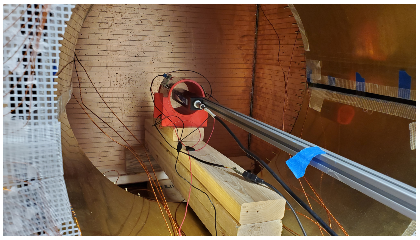
Figure 9: Dr. Crawford's auto-mapper system.

The auto-mapper operating program works by reading from a .txt file specified by the user that contains the list of coordinates where the mapper is to measure and record the B-field. The control program then writes the B-field component measurements and the coordinates at which they were taken to a user specified .txt file.

All field measurements of the coil were taken while the coil was placed inside a Mu-metal shield. Mu-metal is a nickle-iron alloy with a very high permeability.



(a) Dr. Crawford's Mu-metal magnetic shield.



(b) Coil placed in the center of Mu-metal shield.

This was done to reduce interference from external B-field sources that could cause the collected B-field data to contain irregularities that could not be corrected for.

6.2 Mapping the B-Field

To verify that all twenty-one sub-coils were functioning properly, I took B-field maps with full 120 mA current running through only one sub-coil at a time, while using the same grid of points each time. I also took two mappings with a different grid of points: one with all of the sub-coils in a single flex PCB set to full current, and the other with all sub-coils set to full current.

For each of these mappings, I took an additional mapping field directly beforehand with no current running through the coil at all. The reason for this is that at all times and in all places there is a locally uniform and relatively stable B-field produced by the earth. By mapping this ambient B-field and subtracting it from the raw coil-generated B-field, I isolated the B-field generated by only the current in the coil.

7 Simulating Coil and B-Field

With the coils component B-fields mapped, the next step to verify the results was to simulate what the expected B-field should be using the Bio-Savart Law, Python, and Numpy; but before I could do that, I needed twenty-one lists of coordinates to represent the twenty-one sub-coils.

7.1 Coil Geometry

To generate these lists I first made measurements of several important parameters of the flex PCB's geometry, specifically total PCB length, length of solder pads, amplitude of sinusoidal section, and mid-point separation between copper traces. Then, I created the list "arc" to represent the total length of the PCB in millimeters. Using a "for" loop, I then created three lists to represent the path of the copper trace as a function of the length of the PCB. I had now simulated one of the flattened out copper traces.

To simulate the actual coil, I used a nested "for" loop to wrap the traces into a cylindrical shape. The outer loop represented the three flex PCBs, with each iteration rotating the simulated flex PCB by 120° around the y-axis. The inner loop represented the fourteen windings in each flex PCB, with each iteration offsetting the winding by five millimeters in the y direction.

At this point I had three Numpy arrays "x", "y", and "z" each containing forty-two lists representing the x, y, and z coordinates of each winding respectively. To combine these into a list of lists of coordinates, I used the Numpy.stack method and assigned the resulting list of forty-two arrays to "coils". I had now simulated the geometry of my coil.

7.2 Biot-Savart

To simulate the B-field produced by my coil, I wrote a Python function to calculate the B-field at a given grid of points for a given list of coil geometries and currents by approximating the coil as a collection of small straight wires, applying a non-general form of the Biot-Savart Law,

$$\mathbf{B} = \frac{\mu_0}{4\pi} \oint' \frac{\mathbf{v}\, dq' \times \mathbf{r}}{r^3} \approx \sum_i (\Delta \mathbf{B} = \frac{\mu_0}{4\pi} \frac{I\, \Delta \times \mathbf{r}_0}{r_0^3})_i \tag{1}$$

$$\Delta \mathbf{B} = \frac{\mu_0 I}{4\pi} \frac{(\mathbf{r}_- \times \mathbf{r}_+)\,(\mathbf{r}_- + \mathbf{r}_+)}{r_-\, r_+\,(r_- r_+ + \mathbf{r}_- \cdot \mathbf{r}_+)} \tag{2}$$

$$\mathbf{B} \approx \sum_i (\frac{\mu_0 I}{4\pi} \frac{(\mathbf{r}_- \times \mathbf{r}_+)\,(\mathbf{r}_- + \mathbf{r}_+)}{r_-\, r_+\,(r_- r_+ + \mathbf{r}_- \cdot \mathbf{r}_+)})_i \tag{3}$$

to these wire sections, and summing the B-field contributions to find the total B-field.[1] It was important when designing this function to vectorize as many of the calculations as possible and to minimize "for loops" to minimize function runtime.

I was now able to pass a grid of points, a selection of sub-coils, and the corresponding current settings to the function, and the output would be the grid of points and the vector components of the simulated B-field at those points.

8 Processing the Data

Because in the process of mapping the B-field produced by the coil it was not possible to achieve perfect alignment between the coil and the auto-mapper system, to get an accurate idea of how well the coil performs when compared to the simulated coil, which is the idealized result, it is necessary to correct for the small offsets and rotations between the coil and auto-mapper system. I wrote three Python functions to accomplish this.

The first was very simple and only took a list of points, applied a translation and a rotation using Euler Angles, and returned the transformed points.

The second function generated the value that would be minimized to find the correct transformation. It would take a list of nine parameters: rotations about the x-, y-, and z-axes, translations along x,y, and z, and then rotations around each axis again but now applied to the measured vectors instead of the points of measurement. The function would also take multiple mapping measurement sets and lists of active coils and their respective current settings. Using this data, for each mapping the function would apply the transformation using the given parameters, calculate field generate by the difference between the transformed measured field and the simulated field, and then calculate the RMS of the magnitudes of those difference field vectors. Finally the function sums all of the RMSs from all of the mapping datasets given it and returns the value.

The final function, and the one that calculates the best minimizing parameters, simply calls a minimization function with the second function being the thing to be minimized. Once the minimization function finds the best fit parameters, it then applies those rotations and translations to the mapping data sets given it and returns both the values of the bast fit parameters and the now corrected data sets.

Originally, I tried to minimize using a function that only calculated the best fit for a data set from a single mapping, but I found that this caused the minimizing function to greatly over fit the data. I corrected for this by having the minimizing function find the best fit parameters for a set of mapping data sets, and this seems to have been successful in correcting the issue.

9 Results

Now with both mappings of the coil B-fields and simulations of the expected B-fields, a comparison of the two can be made.

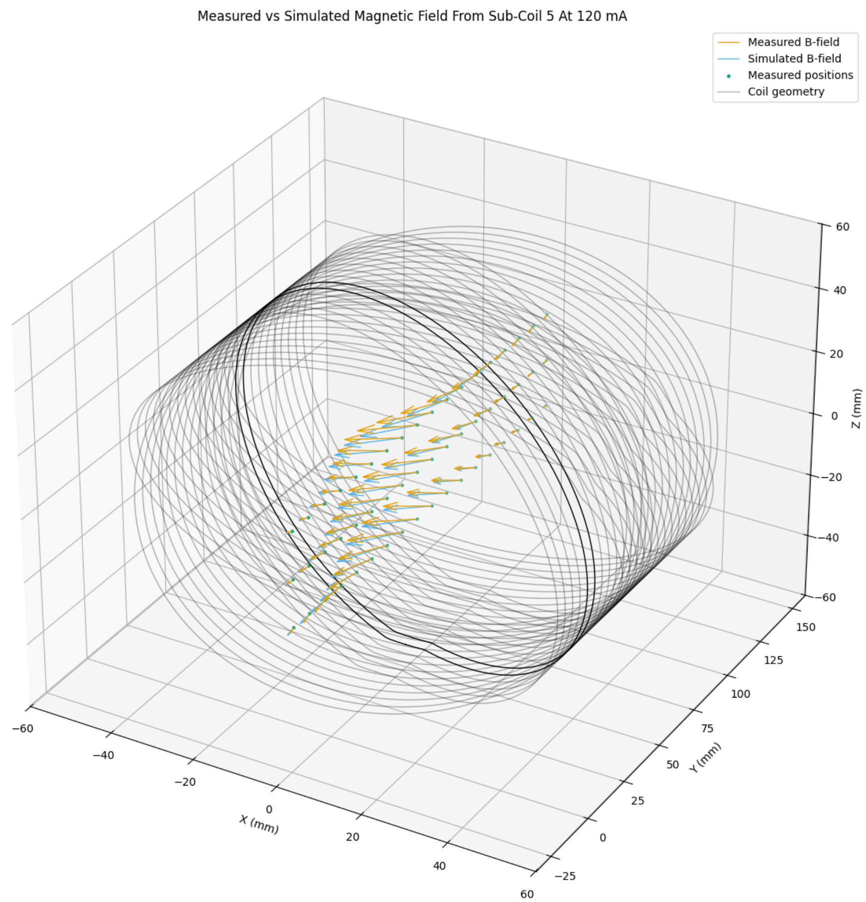


Figure 11

In figure: 11 is shown one of the twenty-one sub-coil full current mappings plotted with the expected B-field for full current on that sub-coil. While the measured and simulated B-field mappings are not perfectly aligned, this is to be expected due to several factors, such as the simulated coil not being a one-to-one replication and possible misalignment of the two mappings coordinates systems. While not shown here to avoid excessive figures, the other twenty mappings display similar levels of agreement.

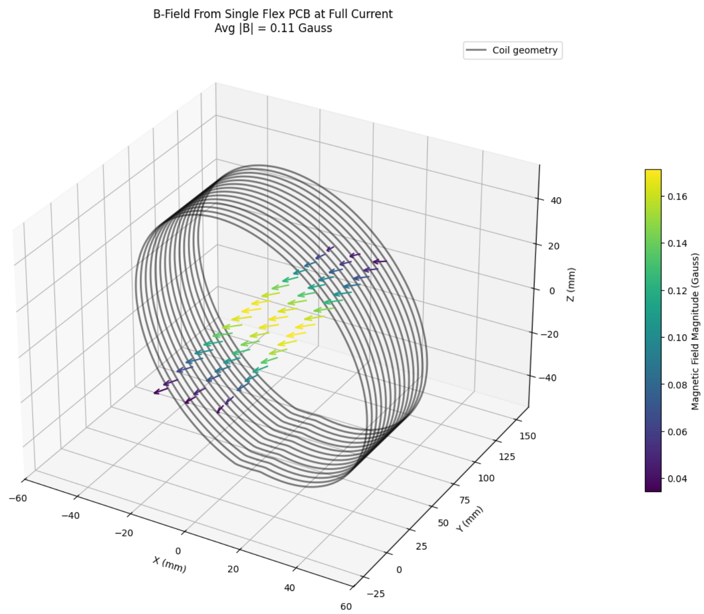


Figure 12

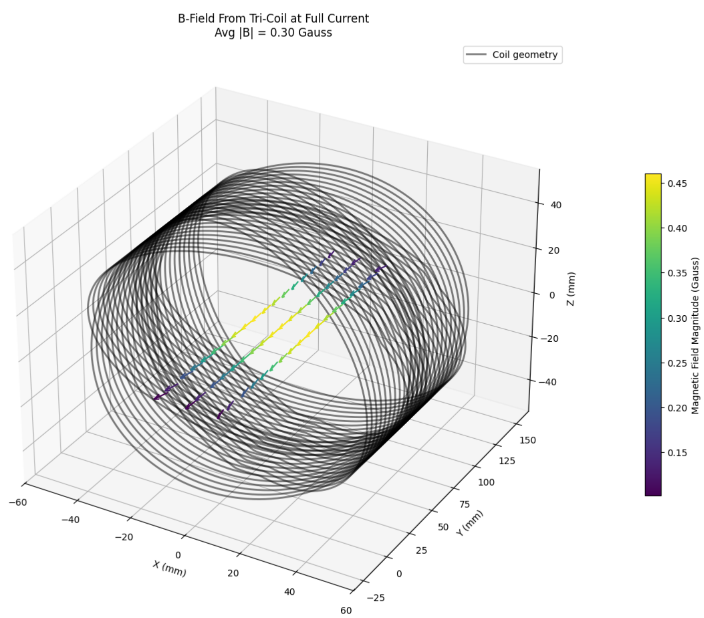


Figure 13

In figures: 12 and 13 are shown two separate measured B-field mappings with B-field strength indicated by the color gradient. The first figure shows the B-field generated when all of the sub-coils on one of the flex PCBs are set to full current. We can see that the B-field is somewhat uniform; however, the strength of the B-field drops off as the distance from the center increases and the orientation of the B-field begins to drift as well. In the second figure, we see the same effect happening to the strength of the B-field; but because this mapping is of the B-field generated when all sub-coils are set to full current, the orientation of the B-field near the axis of the coil remains uniform. This is because all of the radial components of the B-field are being canceled out.

10 Conclusion

Based on the comparison of the measured B-field mappings to the simulated mappings, I am confident that the coil is in general working as intended. Before I can make a more precise verification, I will first need to apply the fitting functions that I have written to the data and I will need to improve the design of the coil geometry simulation. By using the fitting functions, I should be able to minimize the impact of whatever translational and rotational misalignment there happen to be between the recorded positions of the measured B-field maps and the actual position of the B-field mappings. There were several features of the coil that I have not captured in my current coil simulation, such as the increasing diameter and increasing solder pad gaps of the layered flex PCBs. These two factors will need to be corrected for before statistical analysis of the results can begin.

11 Next Steps

For the next phase of this research and development project I will develop an algorithm for the coil code to increase field uniformity and add magnetic sensors to the control circuit boards to correct for external field fluctuations. Using my simulation of the coil, I will solve for the current distribution along the length of the coil that minimizes unwanted magnetic field deviations for any selected magnetic field settings. I will then build this information into the coil operating code, making the correction process automatic. I will also add magnetic sensor chips to the circuitry of the coil, which will monitor the local magnetic field. This information will be actively fed to the coil microcontroller and used to modify the coil's current settings such that fluctuations and changes in the external magnetic field will be canceled out. I will also adapt the design to provide high resolution tuning of the magnitude and direction of the solenoid field for the Project 8 experiment, which has a field uniformity requirement of 100 parts per billion. This experiment will measure the mass of the electron neutrino or place a stringent limit on the mass less than 40 MeV, which is one of the highest priority unsolved problems in nuclear and particle physics.

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